

2. Answer background questions, and upload them to your github (5 pts)

1. Which packages are available for ML? Describe the pros and cons and document the availability.
2. What is ChEMBL? How do you access it?
3. What is machine learning, and how does it differ from traditional programming?
4. What are the key concepts and techniques in machine learning?
5. What are the different types of machine learning algorithms?
6. What are the common applications of machine learning?
7. How do you evaluate the performance of a machine learning model?
8. How do you prepare data for use in a machine learning model?
9. What are some common challenges in machine learning, and how can they be addressed?
10. What are some resources and tools available to help you learn and practice machine learning?

3. answer questions and document the important steps of the process (5 pts)

1. What is in the training set, how big is it?
2. What modifications do you need to do to the data set to perform the tutorial.
3. What is a test set? Any other types of set?
4. Before starting describe with 1-2 sentences, in your own words, what is done in each of the cells.

Read the documentation: <https://hpc-unibe-ch.github.io/>.

1. What is Ubelix?
2. How do you gain access?
3. How do you submit a job?
4. Who can have access?
5. What resources are available there?
6. **Reuse the code from the tutorial to run the relevant part**

(training) on Ubelix. Document how you do the code modification and transfer. (5 pts)

8. Run the training step on Ubelix (10 pts)

Provide the submission script and the code you run there.

9. To obtain full points (use GPT to help you): Easy mode, choose a different target, download active compounds from ChEMBL and train with this. Hard mode, change AI model to SVM and/or RF (random forest) or other. Compare the performance. (10 pts)
